

No.	Name	Description
EK16	Low fire hold or warming limit active	0 - Measured value above warming limit, burner free to modulate. 1 - Measured value below warming limit, burner held at low fire if the control is in auto or sequencing mode.
EK18	Boiler status	0 - Burner will not fire because the temperature / pressure measured value has exceeded the high control value. 1 - Burner will fire because the temperature / pressure measured value has fallen below the low control value.
EK19	Confirm to adjust	0 - Control not in commission mode or control is in commission mode and drives moving to setpoints. 1 - Control in commission mode and drives may be adjusted using the UP/DOWN keys.
EK20	Drive moved	0 - Control not in commission mode or control is in commission mode and drives have not been moved using the UP/DOWN keys. 1 - Control in commission mode and drives have been moved using the UP/DOWN keys.
EK21	Positions proved	0 - Drives have stopped moving, ready for position prove test. 1 - Ready for position prove test but drives still moving.
EK22	Fault alarm	0 - No alarms (faults) present. 1 - Any Un-muted fault alarm present
EK23	Oxygen Trim Enable	0 - Oxygen trim off or not working. 1 - Oxygen trim on and working. Note: EK23 = 1 IF EK46 = 0
EK25	CAN_TX Failures	A count of the number of times the CANbus controller has failed to send a message since power-up. This should be zero unless there has been a problem.
EK26	Commission status	Current commissioning mode, where: 0 - Normal run mode. 1 - Adjust ratio mode. 2 - Commission ratio mode.



No.	Name	Description
EK27	Commission setpoint	The current setpoint being commissioned. 0 - Closed setpoint. 1 - Purge setpoint. 2 - Ignition setpoint. 3 - Low fire setpoint. 4 - 24 - profile setpoints
EK28	Commission Setpoints entered	The number of setpoints that have been successfully entered during this commission ratio session.
EK29	Modulation rate (%)	The current modulation rate of the burner. 0 = Low Fire 100 = High fire
EK30	Burner Status	Status of the start-up sequence. See section 4.2
EK31	Fuel Profile Selected	The currently selected fuel profile.
EK32	Number of commission setpoints	The number of setpoints entered for the current selected profile: 0 - No setpoints entered. 1 - Close setpoint only. 2 - Close and purge setpoints. 3 - Close, purge and ignition. 4 - 24 - profile setpoints.
EK33	Modulation mode	0 - Auto mode. +1 - Manual from keyboard / TSD +2 - Low fire hold from keyboard / TSD +4 - Local 1. +8 - Local 2. +12 - OFF from Display Ex.: 10 would be local 2, LFH. 4 would be local1, auto mode.
EK36	Nearest setpoint	The number of the profile setpoint which is nearest to the current modulation position.
EK37	Voltage reference error (V)	The error between the measured voltage reference and the calibrated value has been exceeded for 3 seconds. Typically this value will be < 0.05v and F19 will occur if it is > 0.12v. If the value is high, ensure none of the low voltage inputs are above 5 volts
EK38	RS485 Communications status	Toggles between 1 and 0 when RS485 is active on the local interface (not the Modbus interface).



No.	Name	Description
EK39	Fuel Swap Status	If non zero, the control is in the process of performing a fuel profile swap (without turning the burner off).
EK40	Shutdown setpoint	The nearest setpoint (EK36) when the burner last locked out.
EK42	Adjust ratio counter	The number of times adjust ratio mode has been used.
EK43	Commission ratio counter	The number of times commission ratio mode has been used.
EK44	Oxygen measured value	The current flue oxygen value as measured by the ETC Oxygen probe (if fitted).
EK45	Oxygen probe status	0 – Probe OK +1 - Internal fault. +2 - Internal fault. +4 - Heater fault. Causes F50. +8 - Cell temperature out of range. +16 - Cell millivolts out of range. +32 - CANbus error. +64 - Probe calibrating in reference gas. See option parameter 30.6. +128 - Probe calibrating in air. See option parameter 30.6.
EK46	Oxygen trim status	0 - O2 trim working, or +1 – no oxygen interface connected. +2 – no oxygen probe serial number entered, or option 30.5 not 1, or option 30.8 not 0 or trim not selected via aux input or trim not selected via serial comms +4 – O2 setpoints or flow values incorrect +8 - Option 20.6 not 0 +16- Not modulating +32- In commission mode +64- Probe faulty (see EK45)
EK47	Oxygen setpoint (%)	Current oxygen setpoint. This is only available if trim is enabled and working.
EK48	Excess air (%)	Excess combustion air at the current firing position.
EK49	Oxygen error (%)	Error between O2 measured value and O2 setpoint.



No.	Name	Description
EK50	Trim deviation (% flow)	Deviation in air flow imposed by trim (-25 to +25%)
EK55	Main PCB issue	The issue number of the main circuit board that the firmware is expecting. This may be different to the actual PCB issue if they are compatible.
EK56	Software issue.	The current revision of the main product firmware.
EK58	CPU serial number (low 4 digits)	The CPU board serial number.
EK59	Drives at setpoint	Represents which drives are currently at their setpoints, where: 0 – All drives are at their setpoints and are not moving. Non- zero – One or more drives are not at their setpoint.
EK60	Digital (relay) outputs on.	The combination of digital outputs currently switched on. +1: Digital Output 1 ON. Display, PR3 (COM), PR1 (NO), PR2 (NC). Low voltage or line voltage. +2: Digital Output 2 ON. Display, PR7 (COM), PR5 (NO), PR6 (NC). Low voltage or line voltage. +4: Digital Output 3 ON. Display, PR7 (COM), PR9 (NO), PR8 (NC). Low voltage or line voltage. +8: Digital Output 4 ON. PPC6000, PE4 (line voltage out). +16: Digital Output 5 ON. Not implemented. +32: Digital Output 6 ON. Not implemented. +64: Digital Output 7 ON. Daughterboard, PZ15 – PZ16. LOW VOLTAGE AND CURRENT ONLY. +128: Digital Output 8 ON. Daughterboard, PZ17 – PZ17. Not implemented.
EK61	Analogue/Digital Input 1. PPC6000 Terminal PA5.	The raw ADC counts from analogue/digital input 1. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts

No.	Name	Description
EK62	Analogue/Digital Input 2. PPC6000 Terminal PA6.	The raw ADC counts from analogue/digital input 2. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts
EK63	Analogue/Digital Input 3. PPC6000 Terminal PA7.	The raw ADC counts from analogue/digital input 3. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts
EK64	Analogue/Digital Input 4. PPC6000 Terminal PA8	The raw ADC counts from analogue/digital input 4. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts
EK65	Analogue Input 5. PPC6000 Terminal PA12 (also remote setpoint).	The raw ADC counts from analogue input 5. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts
EK66	Analogue Input 6. PPC6000 Terminal PB9 (also boiler measured value).	The raw ADC counts from analogue input 6. 0 to 1023 for 0 to 5.00 volts. 4mA = 180 counts 20mA = 900 counts
EK67	Analogue input 7. Daughterboard terminal PZ12 (-) PZ13 (+). Also VSD1.	The ADC counts from analogue input 7. 0 to 999 for 4 to 20mA.
EK68	Analogue input 8. Daughterboard terminal PZ13 (-) PZ14 (+). Also VSD2.	The ADC counts from analogue input 8. 0 to 999 for 4 to 20mA.
Note for EK67 and EK 68: Terminal PZ13 alternates $\pm$ to detect any direct connection between these two inputs.		
EK69	Daughter board Frequency input 1 (PZ7). 0 to 12 volts.	Gives the measured frequency in Hz on this input. If the input is not changing, this value will be 0 for OFF (open circuit) and 1 for ON.
EK70	Daughter board Frequency input 2 (PZ9). 0 to 12 volts.	Gives the measured frequency in Hz on this input. If the input is not changing, this value will be 0 for OFF (open circuit) and 1 for ON.



No.	Name	Description
EK71	Daughter board Frequency input 3 (PZ11). 0 to 12 volts.	Gives the measured frequency in Hz on this input. If the input is not changing, this value will be 0 for OFF (open circuit) and 1 for ON.
EK72	Oxygen probe cell temperature.	The temperature of the zirconia cell inside the ETC oxygen probe, if fitted. This value should be very close to 650°C for accurate operation.
EK73	Ambient temperature.	The temperature measured by the inlet temperature sensor, if fitted. Units are °C.
EK74	CPU utilisation (%).	The percentage utilisation of the CPU. Should be less than 95% at all times.
EK75	Second O2 level (%)	The measured flue oxygen level as measured by the second oxygen probe interface. (%)
EK76	Second probe status	The same as EK45, but for the second oxygen probe.
EK77	Program size.	This is the length of the currently running user / manufacturers program. If a user program is selected, this should match the program length given by the 'abacus' software.
EK78	Second cell temp (°C)	The internal zirconia cell temperature of the second oxygen probe, if fitted.
EK79	Second flue temp (°C)	The flue temperature, as measured by the second oxygen probe, if fitted.
EK80 - 89	Drive Error values for drives 0 to 9.	The current error value for drives 0 to 9. These values will freeze when the control performs a safety shutdown, so it may be possible to look at these values after a lockout to help determine the cause.
EK90	Burner cycles	The number of times the burner has attempted to start.
EK91	Burner Lockouts	The number of times the burner has locked out.
EK92	Commission data backup verification progress	Description This counts up as the backup is verified when the burner is running. If a difference between the actual commissioning data and the backup is found, it will go to zero. A value of 999 indicates that the entire backup has been verified so all commissioning data are backed up in the display.

No.	Name	Description
EK93	Flow value	The instantaneous calculated air or fuel flow value, as used by the oxygen trim function. If flow values have been entered during commissioning for the current profile, this value will show the flow rate that corresponds to the current modulation rate. At high fire, the value will be 99.9%. At low fire, it will be this divided by the turn down of the boiler. If fuel trim is active, this will track air flow. If air trim is active, this will track fuel flow.
EK94	Reserved	N/A
EK95- EK97	Spare	N/A
EK98	Combustion air pressure / error (added with firmware version 1.219) (EK56)	If a combustion air pressure sensor is used (in addition to the main air pressure switch), in commission modes, this will give the measured combustion air pressure. In run mode, this will give the combustion air pressure error which is the difference between the measured and expected air pressure.
EK99	Spare	N/A
EK100	Firmware type PT22...	This shows the last four digits of the firmware part number for the PPC6000 software. It is used for product variant identification.



- The product allows for customization of various non-safety critical functions including the modulation control. The engineers' keys shown below relate to the default modulation control function programmed in the control at the factory. To verify this has not been replaced by an application specific function check with the equipment supplier and / or check option parameter 00.4, if available.

NOTE: EK101-200 (shown in next table) are controlled by the loaded modulation program. The values given below are for the latest (version 40 (EK200)) production version but user programs may change / add to these.

No.	Name	Description
EK101	PID1 Active	If the value is 1, Setpoint / PID set 1 is active.
EK102	PID2 Active	If the value is 1, Setpoint / PID set 2 is active.



No.	Name	Description
EK103	Control Limits Active	If the value is 1, the control limits are active and will turn the burner on and off as the load dictates. The low and high limits are shown on EK153 and 154. If the control limits are holding the burner off, EK18 will be zero.
EK104	User modulation mode	This will normally be zero. A custom modulation program may change this value if it is modifying the modulation rate.
EK105	PID not required	If this value is 1, the internal modulation PID is not running. This may be because the burner is off, in commission / manual mode or for another reason.
EK106	Burner Firing	If this value is 1, the burner is firing (fuel valves open), or it is in post-purge.
EK107	Warming Limit Active	If this value is 1, the warming limit function (see option parameter 23.0) is holding the burner at low fire (status 15).
EK108	Tracking Active	If this value is 1, the remote tracking function is active. See option parameter 20.7. The AUTO modulation rate will come from analog input 5.
EK109	Remote Setpoint 1 Active	If this value is 1, the remote setpoint 1 function is active. See option parameter 20.7. The value of Setpoint 1 will come from analog input 5.
EK110	Control limit exceeded.	
EK111 to EK114.	Not used by the standard manufacturers program. May be used by a custom program.	N/A
EK115	Comms modulation active.	
EK116 to EK132.	Not used by the standard manufacturers program. May be used by a custom program.	N/A
EK133	SP2 selected remotely	If this value = 1, the second PID modulation setpoint (SP2) has been selected remotely via serial communications (including sequencing and Modbus).
EK134	Boiler sequencing communications good	If this value = 1, the control is receiving setpoint selection requests via serial communications (including Modbus and boiler sequencing). This usually indicates that boiler sequencing is working."
EK135	AUX input in use	This value = 1 if option parameter 20.7 (analog input 5 function) is non-zero.

No.	Name	Description
EK136	Lag on temperature active	This value = 1 if the control is currently banking based on a temperature setpoint for SP2
EK137	Remote lead selected	This value = 1 if the another boiler has been selected to be lead boiler instead of this one.
EK138	SP1 selected remotely	If this value = 1, the first PID modulation setpoint (SP1) has been selected remotely via serial communications (including sequencing and Modbus).
EK139	preload lag status	Used internally to keep the same boilers firing when the lead boiler changes.
EK140 to EK 150	Not used by the standard manufacturers program. May be used by a custom program	N/A
EK151	Setpoint 1 control value	The actual value of setpoint 1 being applied.
EK152	Setpoint 2 control value	The actual value of setpoint 2 being applied.
EK153	Low control limit	The actual value of the low control limit (cut-in point) being applied.
EK154	High control limit	The actual value of the High control limit (cut-out point) being applied.
EK155	Spare	N/A
EK156	Measured Value	The current boiler measured value (actual value) shown with increased precision.
EK157	AUTO modulation rate	The modulation rate that the boiler will fire to if in AUTO mode (providing option parameter 15.0 is not zero or 1).
EK158	Not used by the standard manufacturers program. May be used by a custom program.	N/A
EK159	Banking water temperature	Actual water temperature measured in °C or °F
EK160 to EK199.	Not used by the standard manufacturers program. May be used by a custom program.	N/A
EK200	Prog:	Currently running manufacturers / custom (user) modulation program name and revision number (if applicable).

*** EK101-200 are controlled by the loaded modulation program. The values given below are for the latest (version 40) production version but user programs may change / add to these.**



6.7 Troubleshooting

6.7.1 Display / General

Problem	Possible cause	Suggested action
Display shows it's serial number but not data from control unit	Display CAN bus wired incorrectly. PPC6000 not running.	Check wiring of CAN bus (PT3 and PT4). Check PPC6000 low voltage electronics fuse FS2. If blown, investigate all low voltage external wiring and replace fuse, if necessary, with a new one of the correct type and rating.
Display will not light at all.	24VAC supply to display missing. PPC6000 not running.	Check for 24VAC on PT1 and PT2. Check fuse FS1. If blown, investigate all high and low voltage external wiring and replace fuse, if necessary, with a new one of the correct type and rating.
Measured value incorrect.	Wrong sensor voltage. Wrong sensor input type. Incorrect sensor wiring. Wrong zero or span.	Check SENS SUPP link (see 2.5.3). Check SENS IN link (see 2.5.3). Check wiring to terminals PB8 - 10. Check option parameters 15.0, 15.1, 15.2.
Hours run shows '---'.	No profile selected.	Select a profile.
Modulation rate is 1%	Burner not modulating.	Wait for burner to finish start-up sequence. Check PID proportional band – must be greater than zero to modulate.
Burner status is flashing.	Control in non-volatile lockout mode.	Burner status before shutdown displayed.
SEE ALSO – Operation in Local Mode		

Operation in Local Mode

NOTE: This function is not available on displays manufactured after September 2007

Since a common PPC6000 display can be connected up to as many as sixteen PPC6000's at once, a common display must have the desired PPC6000's address set in order to communicate. This is done by entering "local mode" at the display and scrolling up or down to the desired address and exiting local mode. If communications to a display is lost for approximately two seconds, the display enters local mode. If communications is reestablished the display will revert to normal operation.

To enter local mode, press both the left and right arrow keys simultaneously. The display will show the following;

TOP LINE- The address (unit) number it is connected to. **N/C indicates** the PPC addressed is **not present**. Set to match PPC6000 (default is zero).

BOTTOM LINE- Alternates between serial number and software version and the fault logging and commissioning backup status. "FL" and "CDBKP" indicates a fault log and backup of the selected address exists.

To exit local mode, press either the "Data" or "COM" key.